

Std - X

Sub - Science

CHAPTER 1

Chemical Reactions and Equations





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Part - 2

Chemical Equations



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Chemical Reactions and Equations

(Part 2) Chemical Equations

Objectives :

- 1) To study the steps of writing chemical equations
- 2) To balance the chemical equations.

Chemical Equations

Definition

'A chemical equation is the symbolic representation of a chemical reaction in the form of symbols and formulae.'

Que : Why is the need of chemical equation?

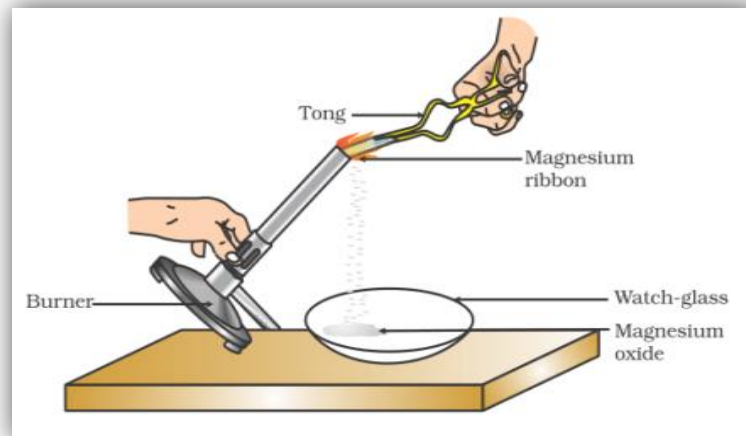
Ans : Sometimes the description of a chemical reaction in a sentence is quite long.

So it can be written in short form using a chemical equation.

Chemical Equations

Example :

When a magnesium ribbon is burnt in oxygen, it gets converted to magnesium oxide.



Chemical equation :

Magnesium + Oxygen → Magnesium Oxide (Word equation)

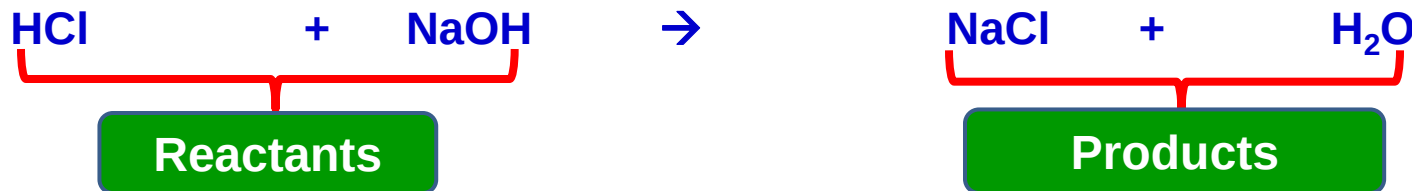
Mg + O → MgO (Symbol equation)

Reactants

Products

Rules for writing Chemical Equations

Example : Consider a chemical equation :



Reactants :

The substances that undergo chemical change in the reaction are called 'reactants'.

The reactants are written on the 'left side' with a plus (+) sign between them.

Arrowhead (→):

The arrowhead Points towards the products, and shows the direction of the reaction.

Products :

The new substances formed during a chemical reaction are called 'products'.

The products are written on the 'right side' with a plus (+) sign between them.

Writing a Chemical Equation

A chemical equation represents a chemical reaction.

Law of conservation of mass states that, “The total mass of elements present in the products of a chemical reaction is equal to the total mass of elements present in the reactants.”



Reactants

Product

- Count and compare the number of atoms of each element on the LHS and RHS of the arrow.
- The number of atoms of elements on both the sides are not the same. Hence, the given chemical equation is unbalanced equation.
- An unbalanced chemical equation in which the ‘reactants’ and ‘products’ are not equal, is called a skeletal chemical equation’ for a chemical reaction.
- So according to the law of conservation of mass, it is important to write the chemical equation by balancing it.

Balanced Chemical Equation

- Remember :
- The number of atoms of each element remains the same before and after a chemical reaction.
- **Example –**
- When zinc granules are added to dil. sulphuric acid, it produces 'zinc sulphate' and hydrogen gas.
- Word equation is written as -

Zinc + Sulphuric acid → Zinc sulphate + Hydrogen

- Chemical equation is written as -



Reactants

Product

Balanced Chemical Equation

Lets see the number of atoms on LHS and RHS are equal or not...



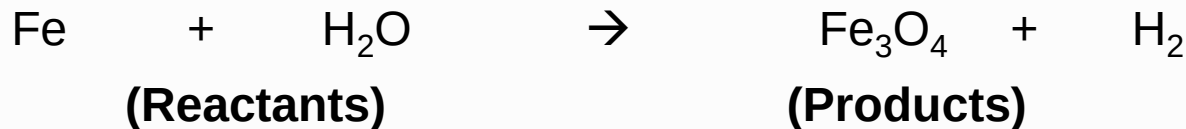
Element	Atoms in Reactant (LHS)	Atoms in products (RHS)
Zn	1	1
H	2	2
S	1	1
O	4	4

So this is a Balanced chemical equation.

Steps to Balance a Chemical Equation

Steps to Balance a Chemical Equation

Balance the following chemical equation :



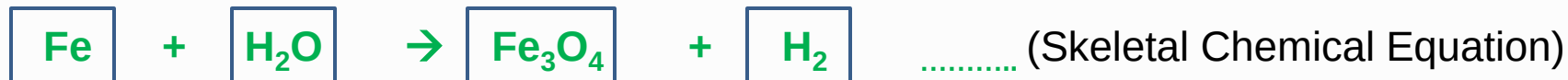
Step – 1 :

Draw boxes around each formula.

Do not change anything inside the boxes while balancing the equation.



Steps to Balance a Chemical Equation



Step – 2 :

List the number of atoms of different elements present in Skeletal equation.

Element	Atoms in Reactant (LHS)	Atoms in products (RHS)
Fe	1	3
H	2	2
O	1	4

So this is a Unbalanced chemical equation.

Steps to Balance a Chemical Equation

Step – 3 :

Start balancing with the compound having maximum number of atoms.

So select Fe_3O_4 and the element oxygen in it.

There are 4 Oxygen atoms on RHS and only 1 Oxygen on LHS.



To balance the oxygen atoms – Multiply oxygen on Reactant (LHS) by 4.

Steps to Balance a Chemical Equation

Step – 4 :

Now balance Fe and H atoms.

First take 'hydrogen' atom.

To equalise 'hydrogen' atoms, make the number of molecules of 'H' as '4' RHS.



Atoms of Hydrogen	Atoms in Reactant (LHS)	Atoms in products (RHS)
1) Initial	8 (in 4 H ₂ O)	2 (in H ₂)
2) To balance	8	2 X 4

Steps to Balance a Chemical Equation

Step – 5 :

Now pick up the third element Fe.



To equalize 'Fe' atoms, we take the 3 atoms of 'Fe' on LHS.

Atoms of Iron	Atoms in Reactant (LHS)	Atoms in products (RHS)
1) Initial	1 (in Fe)	3 (in Fe₃O₄)
2) To balance	1 X 3	3

Steps to Balance a Chemical Equation

Step – 6 :

Finally check the correctness of the balanced equation.



Element	Atoms in Reactant (LHS)	Atoms in products (RHS)
Fe	3	3
H	8	8
O	4	4

The number of atoms of elements on both sides of equation are equal.

Now, the equation is '**balanced**'.

This method of balancing the chemical equation is called '**hit and trial**' method as we make trials to balance the equation by using the smallest 'whole number coefficient'.

Steps to Balance a Chemical Equation

Step – 7 : Writing Symbols of Physical state -

To make the equation more informative, the physical states of the reactants and products are mentioned along with their chemical formulae.

The notations for states are as follows :

	Physical State	Symbol
1	Gaseous	g
2	Liquid	l
3	Aqueous	aq
4	Solid	s

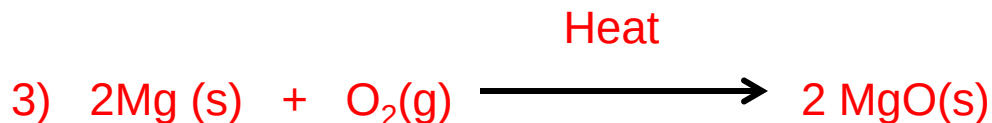
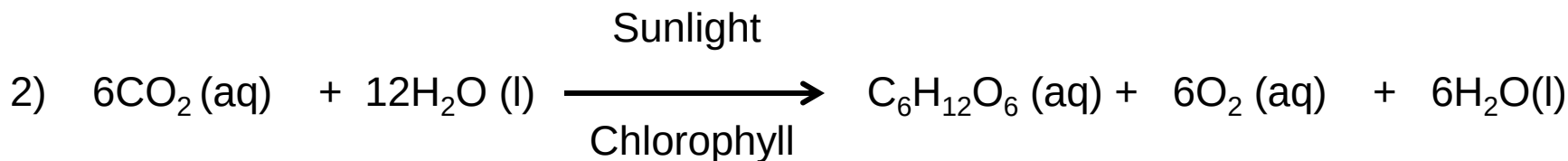


Steps to Balance a Chemical Equation

Step – 8 : Writing reaction conditions –

Sometimes the reaction conditions such as temperature, pressure, catalyst, etc., are indicated for the reaction above and/or below the arrow.

For example :





Home - Work



Q U E S T I O N S

1. Why should a magnesium ribbon be cleaned before burning in air?
2. Write the balanced equation for the following chemical reactions.
 - (i) Hydrogen + Chlorine $\rightarrow$ Hydrogen chloride
 - (ii) Barium chloride + Aluminium sulphate $\rightarrow$ Barium sulphate + Aluminium chloride
 - (iii) Sodium + Water $\rightarrow$ Sodium hydroxide + Hydrogen
3. Write a balanced chemical equation with state symbols for the following reactions.
 - (i) Solutions of barium chloride and sodium sulphate in water react to give insoluble barium sulphate and the solution of sodium chloride.
 - (ii) Sodium hydroxide solution (in water) reacts with hydrochloric acid solution (in water) to produce sodium chloride solution and water.





Home - Work



Q U E S T I O N S

1. Why should a magnesium ribbon be cleaned before burning in air?

- Magnesium is a reactive metal.
- It combines with oxygen in air to form a layer of magnesium oxide on its surface.
- This magnesium oxide layer prevents the burning of the ribbon.
- So as to remove the oxide layer, a magnesium ribbon should be cleaned with a sand paper.



Home - Work



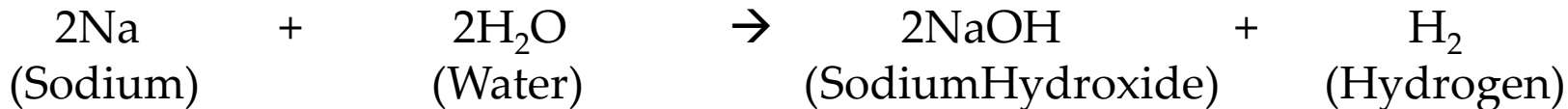
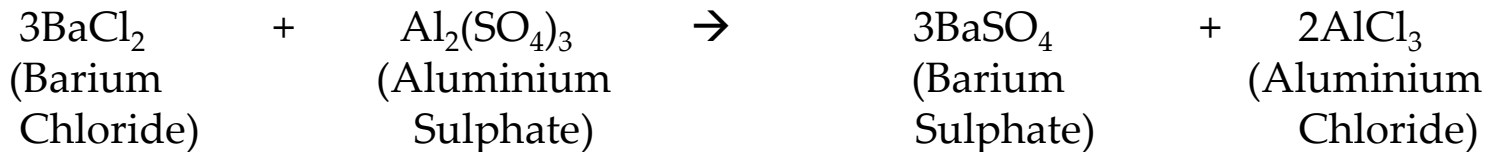
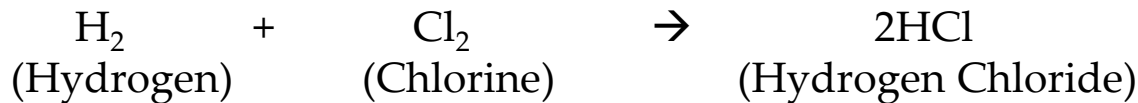
Q U E S T I O N S

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Home - Work

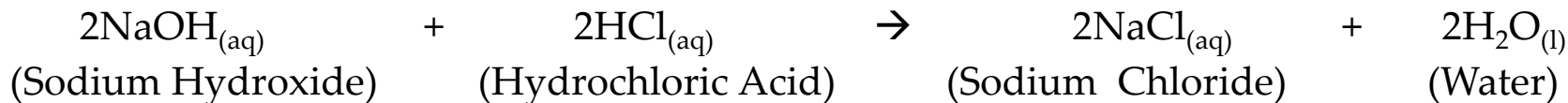


Q U E S T I O N S

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TO BE CONTINUED

